

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES
WEEK 2 | CYBERSECURITY | NETWORKWALKS

Pentester Name	Kishore B (Cybersecurity Intern)
Program/Batch	Networkwalks Cybersecurity Program, Week 02
Date	18 September 2026
Activities	Footprinting & Reconnaissance + Network Scanning
Client/Target	1. networkwalks.com (External Domain) 2. 192.168.56.1 (Authorized Local Network Target)
Permission Secured?	Yes
Platform	Kali Linux
Network Scanner	Zenmap / Nmap

1. Liability Disclaimer

Important: This project was completed for authorized cybersecurity education and practical learning. The techniques documented in this report were only used against systems, applications, and networks where explicit authorization was provided. No unauthorized access or exploitation was performed. Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been secured.

2. Project Overview

This report covers two major cybersecurity activities completed during Week 2 of the Networkwalks Cybersecurity Program: **Footprinting & Reconnaissance using Kali Linux**, and **Network Scanning using Zenmap**.

The footprinting activity focused on collecting publicly available information about the networkwalks.com domain using multiple reconnaissance tools to demonstrate how an attacker maps external infrastructure.

The network scanning activity focused on performing a Ping Scan against an authorized local network target (192.168.56.1) using Zenmap to identify whether the target host was active.

Together, these activities demonstrate the reconnaissance and network discovery stages of a basic penetration testing workflow.

3. Tools Used

Tool	Purpose
Kali Linux	Operating system used for footprinting activities
WHOIS	Domain registration information mapping
WhatWeb	Web technology fingerprinting
Nslookup	Domain-to-IP resolution
Curl	HTTP response header inspection
Wafw00f	Web Application Firewall (WAF) detection
DNSRecon	DNS record enumeration
Zenmap / Nmap	Network discovery and host scanning

4. Activities Performed

4.1 Footprinting & Reconnaissance (networkwalks.com)

Reconnaissance was performed against the `networkwalks.com` domain to collect publicly exposed infrastructure data. The following tools were utilized:

- **WHOIS (Domain Registration Information):** The WHOIS output identified the registrar as **GoDaddy** and the name servers as associated with **HostGator**. The query also revealed the domain creation date, expiry date, and showed that DNSSEC was listed as unsigned.
- **WhatWeb (Web Technology Fingerprinting):** The scan identified the underlying technologies, including the Apache web server, **WordPress 7.1.1**, **WordPress Download Manager 3.3.58**, jQuery, Bootstrap, and Google Tag Manager. Technology and version information can help security teams understand what software is publicly exposed.
- **Nslookup (Domain to IP Resolution):** Using Google's public DNS resolver, the domain name successfully resolved to the IP address `192.232.216.135`.
- **Curl (HTTP Response Header Analysis):** Inspecting the HTTP response headers exposed Apache and WordPress-related information, HTTP security headers, and a direct WordPress REST API reference at `/wp-json/`.

- **Wafw00f (WAF Detection):** The tool identified that the target website is protected by the **ModSecurity (SpiderLabs)** Web Application Firewall. The presence of a WAF indicates that a web application security control is deployed in front of the website.
- **DNSRecon (DNS Enumeration):** The enumeration provided information related to SOA, NS, MX, A, TXT, SPF, and SRV records. The scan also revealed service-related records associated with email discovery.

4.2 Network Discovery with Zenmap

A Ping Scan was executed against an authorized local network target using the Zenmap GUI.

- **Target IP:** 192.168.56.1
- **Scan Profile:** Ping Scan
- **Observation:** The scan successfully completed in approximately 0.72 seconds. The report confirmed *"Host is up. Nmap done: 1 IP address (1 host up) scanned"*. This confirmed that the selected authorized local host was active and reachable during the scan.

5. Findings & Risk Analysis

The following observations were identified during the footprinting and network discovery activities. These are observations from reconnaissance, not confirmed exploitable vulnerabilities.

#	Finding / Observation	Evidence	Potential Impact	Risk
1	Web technology information exposed	WhatWeb	Technology information may assist further security assessment	Low
2	Public IP address identifiable	Nslookup	Provides information about publicly reachable infrastructure	Low
3	Technical HTTP information exposed	Curl	May assist technology fingerprinting	Low
4	WAF technology identifiable	Wafw00f	Reveals information about deployed security architecture	Info
5	DNS infrastructure information exposed	DNSRecon	Can help build an infrastructure profile	Low
6	DNSSEC shown as unsigned	WHOIS	May reduce protection against certain DNS-related risks depending on architecture	Info
7	Authorized local host discovered	Zenmap	Confirms that the selected host was reachable during the scan	Info

6. Security Recommendations

Based on the observations from the activities, the following security practices are recommended to harden the external and internal attack surface:

1. **Review Publicly Exposed Technology Information:** Organizations should periodically review information about their CMS, plugins, frameworks, and web servers that can be identified externally.
2. **Keep Software Updated:** CMS platforms (e.g., WordPress), plugins, and other web technologies should be kept updated and reviewed against relevant security advisories.
3. **Review HTTP Response Headers:** HTTP headers should be reviewed regularly to determine whether unnecessary technical information is being exposed.
4. **Review DNS Records:** DNS records should be periodically reviewed to ensure that only required records and services are publicly exposed.
5. **Maintain WAF Protection:** The detected ModSecurity WAF should remain properly configured, monitored, and maintained.
6. **Review DNS Security:** The organization's DNS security requirements should be reviewed to determine whether additional DNS security controls (such as implementing DNSSEC) are appropriate.
7. **Perform Regular Network Discovery:** Organizations should periodically review their own networks to identify active and authorized devices.
8. **Investigate Unknown Devices:** Unexpected devices discovered during authorized network discovery should be investigated and verified immediately.

7. Conclusion & Lessons Learned

Through this Week 2 project, I learned how reconnaissance and network discovery are used during a cybersecurity assessment. This assessment demonstrated how publicly available information can reveal useful details about an organization's external infrastructure without performing active exploitation.

By utilizing multiple tools, a comprehensive profile of the target was built, ranging from domain registration (WHOIS) and web technology fingerprinting (WhatWeb) to WAF detection (Wafw00f) and DNS enumeration (DNSRecon). Additionally, internal network discovery using Zenmap highlighted how easily active hosts can be identified on a local subnet.

The exercise emphasizes the importance of carefully managing public-facing technical information, properly configuring security headers, and maintaining up-to-date documentation of authorized network devices.

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